README FILE

I am using the JUCE for the MAT500 Project 8.
I am writing this project in Visual Studio Community 2022, on Windows 11 OS. I am submitting this zip file "Project-8". Unzipping this folder you will be having a AudioGraphing.exe file, a “Project-8_d.Namballa.zip" zip file and a README.pdf file.

STEP 1:
Go to this website link
https://juce.com/download/

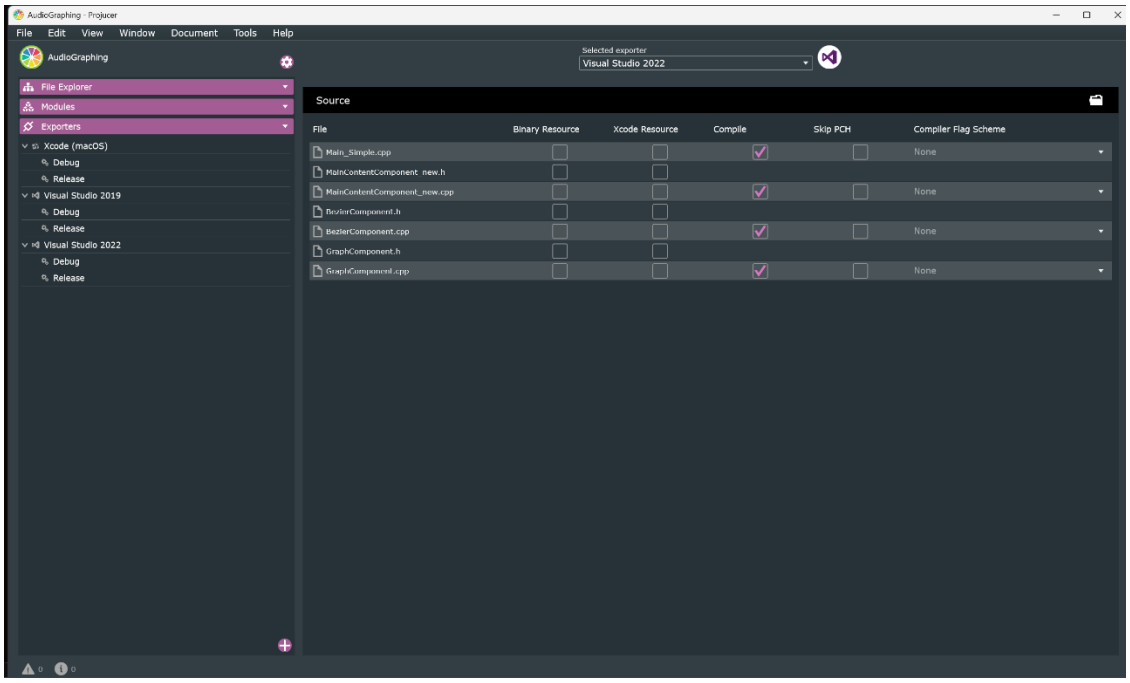
STEP 2:
Download the JUCE and Projucer for Windows. (I am using Windows OS)

STEP 3:
Unzip the downloaded folder juce-8.0.12-windows.zip. Once that is done you will find a folder called "JUCE". Copy and paste this folder in the C drive. (I placed it inside the C Drive)

STEP 4:
The zip folder I submitted is "Project-8”. Unzip this folder and you will be having the “Project-8_d.Namballa.zip" inside it. Unzip this file and copy the unzipped file.

STEP 5:
Open the folder "JUCE" placed inside the C Drive, and you will find a folder called "examples". Copy and paste this folder "Project-8_d. Namballa" from the previous step into the "examples" folder.

STEP 6:
Now, go to the "JUCE" folder in the C Drive and once you open it you will find this file "Projucer.exe". Double clicks on it and it will open the JUCE Editor. Which looks like this

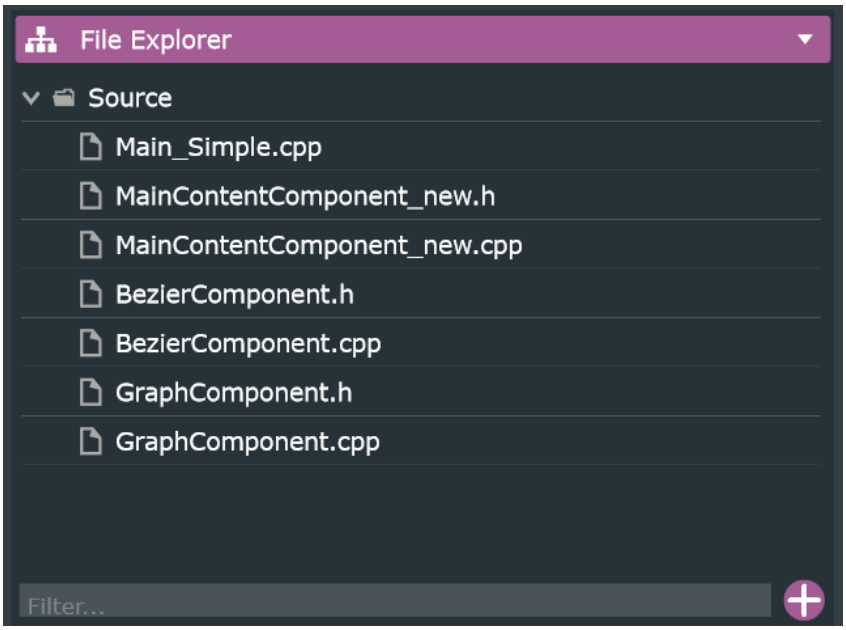


STEP 7:

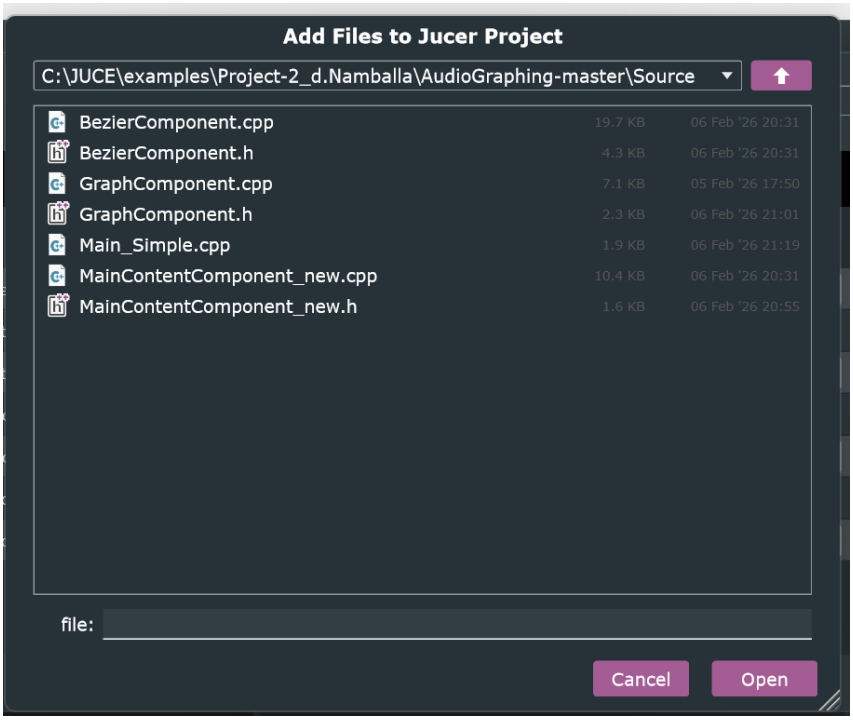
Go to the Files → Open → and Go to the "examples" folder present in the JUCE and open it and inside it you will find this folder called "AudioGraphing-master". Open this folder and you can find this file inside "AudioGraphing.jucer". Double click on this file and it will open this files from the project in the JUCE Editor.

STEP 8:

Check the File Explorer in it and it must contain the following files in the image.

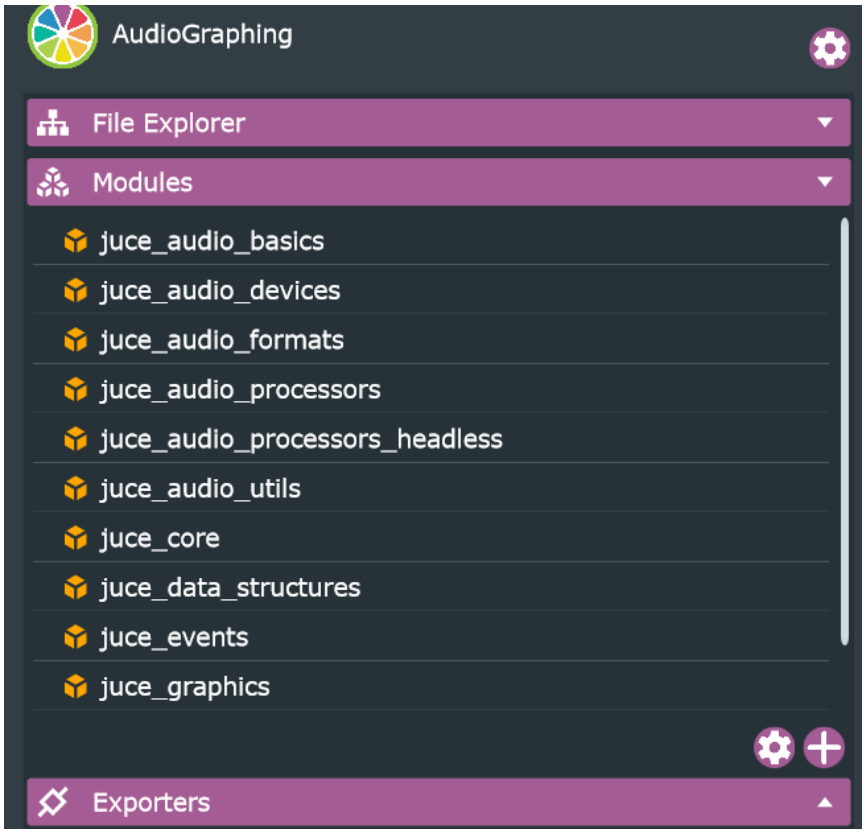


If they are not present then at the bottom of the above image you can see a Filter with a white plus sign in a pick circular button. Click on the circular button with a white plus sign on it and click on the "Add Existing Files". It opens the following window and select the files in it and click Open and it will add to the File Explorer in the JUCE Editor.



STEP 9:

Then go to the next tab called Modules on the left and see of any modules are high lighting as red. If so then click on the pink circular button on the right end and click “add module” and you can see an option called “Re scan”. It will fix it.



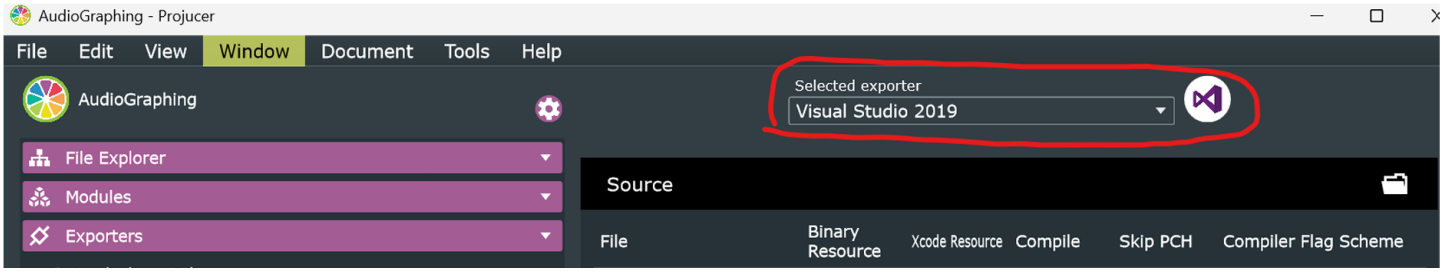
STEP 10:

After that check with the Exporters below the Modules. In the Exporter we will have a list of the IDES that are present on the computer.

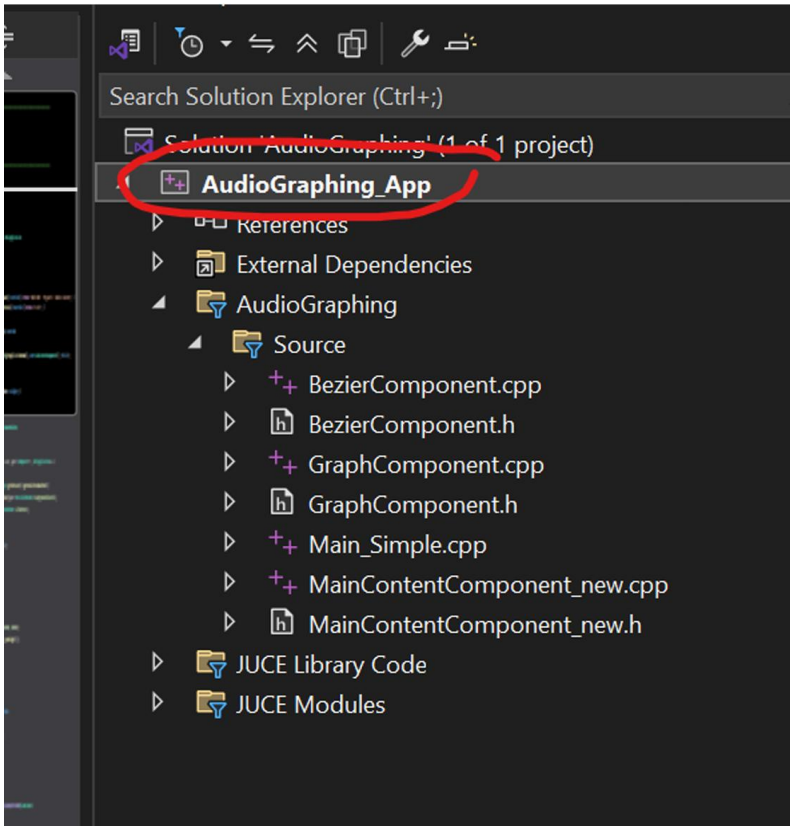


If they are not auto detected, then click on the pink circular button at the right bottom and it opens a window where you can select with IDE you're using. For me I was using Visual Studio Community 2022. Select it from the list whichever you're using.

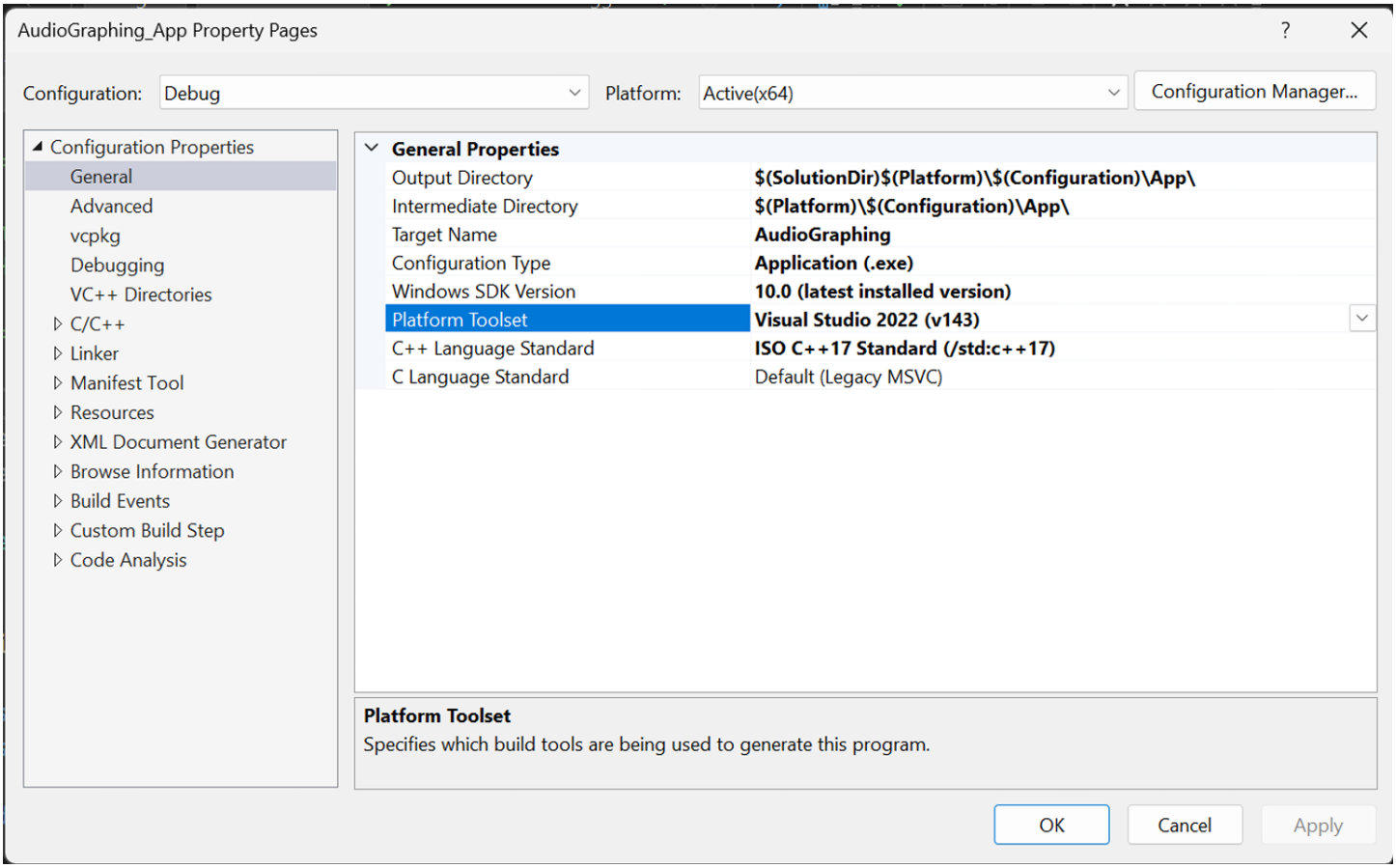
NOTE 1: Initially when I selected my version in the Exporter and select the File on the top left and selecting the "Open in IDE" option it doesn't open. I changed the exporter to the 2019 and repeated by going to File → "Open in IDE" it opened the project in the Visual Studio Community.



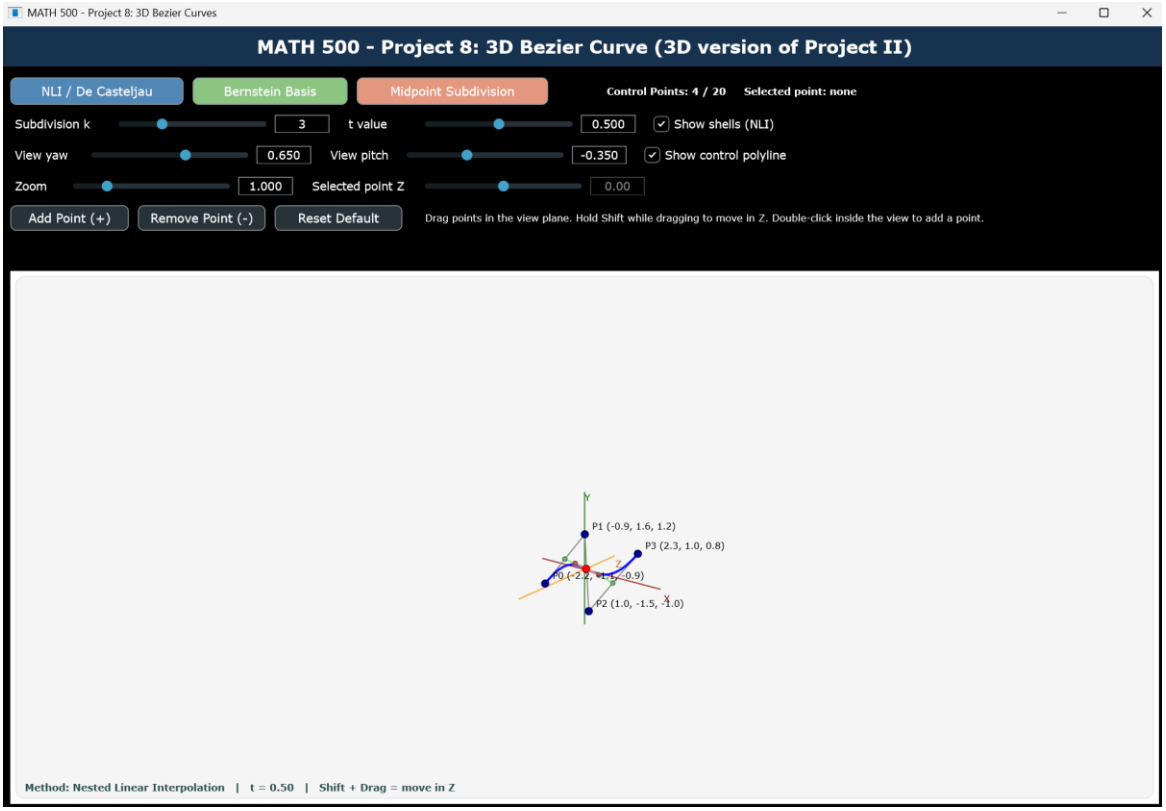
NOTE 2: Once opening the project in the Visual Studio Community and running it if faced with a error related to the v143 or v142 (for me I faced it) I resolved it by going to the highlighted by red in the blow image and right click on it and go to properties.



Which opens the window below go the Configuration Properties → General → Platform Toolset → select the toolset available and then build and run it. It should work.



STEP 11: Then you can run the application in the Visual Studio Community, and it will give you a window like this



Controls Overview

UI Controls – Project 8 (3D Bezier Curve)

1. Control Points (Mouse Interaction)

- **Click & Drag:** Move control points in the X–Y plane
- **Shift + Drag:** Move the selected control point along the Z-axis
- **Double Click:** Add a new control point
- Control points define the shape of the Bezier curve and can be adjusted in real time

2. Camera / View Controls

- **Yaw Slider:** Rotates the view left and right around the curve
- **Pitch Slider:** Rotates the view up and down
- **Zoom Slider:** Zooms the view in and out

These controls change the **viewpoint of the scene**, allowing visualization of the curve in 3D.

3. Curve Parameter (t-Slider)

- Controls the parameter (t € [0,1])
- Moves a point along the curve
- Updates the intermediate **De Casteljau shells** dynamically

4. Curve Evaluation Methods

- Allows switching between different computation methods:
  - **Nested Linear Interpolation (De Casteljau Algorithm)**
  - **Bernstein Polynomial (BB Form)**
  - **Midpoint Subdivision**
- The curve updates instantly when switching methods

5. Visual Elements

- **Control Polyline:** Connects control points
- **Bezier Curve:** Rendered approximation of the curve
- **Shells (NLI mode):** Intermediate interpolation lines for a given (t)